

PROJECT REPORT

ON

KIIT CAMPUS NAVIGATOR USING A* SEARCH

ALGORITHM

Submitted in partial fulfillment for the course
Artificial Intelligence Lab

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ABSTRACT

The KIIT Campus Navigator is a web-based application designed to assist users in finding the shortest path between different locations within the university campus. Large university campuses often create navigation difficulties for new students and visitors, making it challenging to locate various buildings and facilities efficiently.

The system is developed using Python and the Flask framework and utilizes the A* (A-Star) search algorithm to determine the optimal route between selected source and destination points. The campus map is represented as a graph where locations act as nodes and connecting paths act as edges with associated distances. A heuristic function based on geographical coordinates is used to improve the efficiency of the search process.

In addition to the existing text-based search functionality, the system has been enhanced by integrating a voice search feature using Python SpeechRecognition. This allows users to provide input through speech, which is then converted into text and processed by the system. This addition makes the application more interactive and improves ease of use, especially in situations where typing may not be convenient.

The project demonstrates the practical implementation of Artificial Intelligence in solving real-world navigation problems and highlights how additional features like voice-based interaction can further improve user experience in smart campus systems.

INTRODUCTION

The rapid growth of large educational institutions and university campuses has made navigation within campus premises increasingly challenging for students, faculty members, and visitors. New students and visitors often face difficulty in locating different departments, academic blocks, hostels, and administrative offices within a large campus. Traditional navigation methods such as static maps and signboards are often insufficient, leading to confusion and increased time consumption.

With the advancement of technology and Artificial Intelligence, intelligent navigation systems can be developed to provide accurate and efficient pathfinding solutions. AI-based search algorithms are widely used in real-world applications such as GPS navigation, robotics, and game development to determine optimal paths between locations. Implementing such algorithms in a campus environment can significantly improve navigation efficiency and overall user experience.

The KIIT Campus Navigator is an AI-based web application designed to determine the shortest path between various locations within the university campus. The system utilizes the A* (A-Star) search algorithm, which is known for its efficiency in pathfinding and graph traversal. By representing campus locations as nodes and pathways as edges in a graph structure, the system calculates the most optimal route between the selected source and destination.

The project is developed using Python and the Flask web framework, providing an interactive and user-friendly interface. In the earlier version, the system supported only text-based input. In this updated version, a voice search feature has been integrated using Python SpeechRecognition. This allows users to input their queries through speech, making the system more interactive and convenient to use. The addition of voice-based interaction enhances accessibility and improves the overall usability of the system.

Overall, the project demonstrates the practical application of Artificial Intelligence concepts in solving real-world navigation problems and highlights the importance of intelligent systems in modern smart campus environments.

PROBLEM STATEMENT

The rapid expansion of large university campuses often creates difficulty for students, faculty members, and visitors in locating different academic and administrative facilities efficiently. Traditional navigation methods such as static maps and signboards are often insufficient and time-consuming, especially for new users who are unfamiliar with the campus layout.

There is a need for an intelligent system that can provide accurate and efficient navigation within the campus environment. The system should be capable of determining the shortest and most optimal path between different campus locations while considering distance and connectivity.

In addition to route optimization, there is also a growing need to make such systems more user-friendly and accessible. Users may find it inconvenient to type inputs in certain situations, which highlights the importance of incorporating alternative interaction methods such as voice-based input.

Therefore, the problem addressed in this project is to design and develop an AI-based campus navigation system that not only provides optimal pathfinding but also supports intuitive user interaction through both text and voice inputs.

OBJECTIVES

The main objectives of the KIIT Campus Navigator project are:

- To design and develop an AI-based campus navigation system.
- To implement the A* (A-Star) search algorithm for efficient and optimal pathfinding.
- To represent campus locations as nodes and pathways as edges using a graph-based structure.
- To calculate the shortest path between selected source and destination locations.
- To develop a user-friendly web interface using Flask.
- To integrate both text-based and voice-based search functionalities for user input.
- To enhance user experience by enabling hands-free and more accessible interaction.
- To demonstrate the practical application of Artificial Intelligence in solving real-world navigation problems.

SYSTEM ARCHITECTURE

The system architecture of the KIIT Campus Navigator is designed to provide an efficient and user-friendly navigation experience within the university campus. The system follows a client-server architecture where the user interacts with a web interface, and the backend processes the navigation requests using Artificial Intelligence algorithms.

The user can provide input in two ways: by typing the source and destination locations or by using the voice search feature. For voice input, the system uses a microphone to capture the user's speech. This input is processed using the Python SpeechRecognition library, which converts the spoken words into text. The converted text is then automatically used as input for the navigation system.

Once the input is received, the request is sent to the Flask-based backend server. The campus map is represented as a graph consisting of nodes and edges, where each node represents a campus location and each edge represents a path with an associated distance.

The backend implements the A* (A-Star) search algorithm to determine the shortest path between the selected locations. A heuristic function based on geographical coordinates is used to estimate distances and improve the efficiency of the search process.

The frontend interface is implemented using Leaflet.js with OpenStreetMap integration to visually display campus locations and routes. After processing the request, the system returns the optimal path along with distance information, which is then displayed on the map interface.

The inclusion of the voice recognition module enhances the overall system by allowing more natural and convenient interaction. This architecture ensures smooth communication between the user interface, voice processing module, and backend algorithm, resulting in accurate and efficient navigation.

TECHNOLOGIES USED

The KIIT Campus Navigator system is developed using a combination of programming languages, frameworks, and libraries to ensure efficient performance and user-friendly interaction. The main technologies used in this project are:

- **Python** – Used as the core programming language for implementing the backend logic and the A* search algorithm.
- **Flask** – A lightweight web framework used to build the backend server and handle user requests.
- **SpeechRecognition (Python Library)** – Used to implement the voice search feature by converting speech input into text.
- **HTML, CSS, JavaScript** – Used for designing and developing the frontend user interface.
- **Leaflet.js** – A JavaScript library used for interactive map visualization.
- **OpenStreetMap** – Provides the map data for displaying campus locations and routes.
- **Graph Data Structures** – Used to represent campus locations (nodes) and paths (edges) for efficient pathfinding.

These technologies work together to create an interactive and intelligent navigation system that supports both text-based and voice-based user input.

ALGORITHM USED

The KIIT Campus Navigator system uses the A* (A-Star) search algorithm to determine the shortest path between selected campus locations. A* is an informed search algorithm that is widely used in pathfinding and graph traversal problems due to its efficiency and accuracy.

The algorithm works based on an evaluation function given by:

$$f(n) = g(n) + h(n)$$

Here, $g(n)$ represents the actual cost from the starting node to the current node, while $h(n)$ represents the estimated cost from the current node to the destination (heuristic). The function $f(n)$ gives the total estimated cost of the path passing through the current node.

In this project, the campus map is represented as a graph consisting of nodes and edges. Each node represents a campus location, and the edges represent the paths between them along with their respective distances. The heuristic function is calculated using geographical coordinates, which helps estimate the straight-line distance between two locations.

The A* algorithm maintains two lists: an open list containing nodes to be evaluated and a closed list containing nodes that have already been evaluated. At each step, the node with the lowest $f(n)$ value is selected, and its neighboring nodes are explored. This process continues until the destination node is reached.

By combining the actual path cost and heuristic estimation, the A* algorithm efficiently finds the shortest and most optimal route. This makes it highly suitable for applications like campus navigation, where quick and accurate pathfinding is required.

TIME COMPLEXITY

The time complexity of the A* (A-Star) search algorithm depends on the number of nodes and edges present in the graph representing the campus map. In the worst-case scenario, the time complexity of the A* algorithm is similar to that of Dijkstra's algorithm and can be expressed as $O(E \log V)$, where E represents the number of edges and V represents the number of vertices.

The use of a heuristic function in the A* algorithm helps reduce the number of nodes explored during the search process. By guiding the algorithm towards the destination, it avoids unnecessary exploration and improves overall efficiency in most practical cases.

In the context of the campus navigation system, the graph size is limited and predefined, which allows the algorithm to perform efficiently and provide quick results for route calculation.

IMPLEMENTATION

The KIIT Campus Navigator system was implemented using Python for backend processing and Flask as the web framework to handle user requests and responses. The system integrates pathfinding logic, map visualization, and voice recognition to provide an interactive navigation experience.

The campus map was modeled as a graph using appropriate data structures, where each location was defined as a node and the paths between them were represented as edges with associated distances. This graph structure was used as input for the A* search algorithm.

The A* algorithm was implemented in Python using standard data structures such as lists and priority queues to efficiently manage node exploration. The algorithm calculates the shortest path based on both actual distance and heuristic estimation using geographical coordinates.

For handling user interaction, a web interface was developed using HTML, CSS, and JavaScript. The map visualization was implemented using Leaflet.js along with OpenStreetMap, allowing users to view locations and routes interactively.

The system supports both text-based and voice-based input. For voice input, the SpeechRecognition library in Python was used. The system captures audio input through a microphone, processes it using a speech recognition engine, and converts it into text. This text is then automatically used as input for the navigation process.

Flask routes were used to handle user requests, process the input, run the algorithm, and return the results to the frontend. The final output, including the shortest path and distance, is displayed on the map interface for easy understanding.

Overall, the implementation integrates multiple components seamlessly, combining Artificial Intelligence algorithms with web technologies and voice recognition to create a functional and user-friendly campus navigation system.

RESULT AND OUTPUT

The KIIT Campus Navigator system was successfully implemented and tested for various campus locations. The application allows users to select or input a starting point and destination within the campus and calculates the shortest path using the A* (A-Star) search algorithm.

The system supports both text-based and voice-based input. In the case of text input, users can manually enter the source and destination. For voice input, the system captures the user's speech through a microphone, converts it into text using the SpeechRecognition library, and processes it as input for navigation.

The output is displayed on an interactive map interface, where the shortest path between the selected locations is clearly highlighted along with the calculated distance. This makes it easy for users to understand and follow the route.

Example Results:

1. Shortest Path (Campus 6 → Campus 25)

Distance: 0.4 km

The system efficiently calculates the direct route with minimal node exploration.

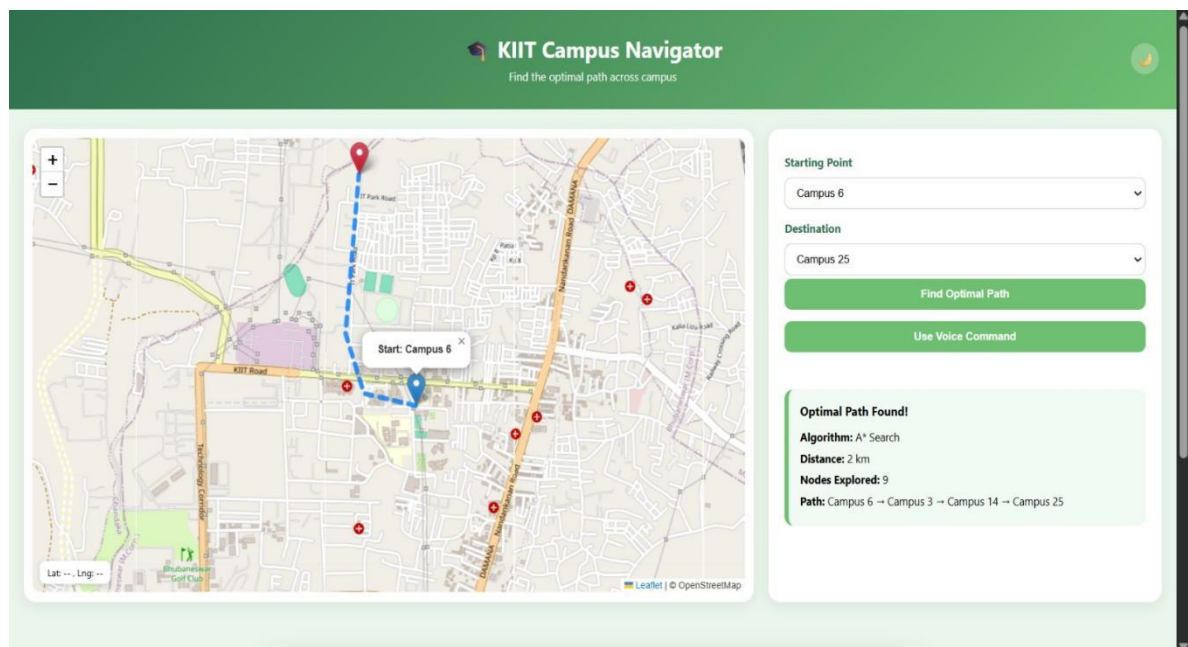


Figure 1: Shortest path from Campus 6 to Campus 25 highlighting multiple nodes explored and the final route selected by the algorithm.

2. Additional Path (Campus 6 → Campus 15)

The system successfully determines an alternative optimal route between different campus locations.

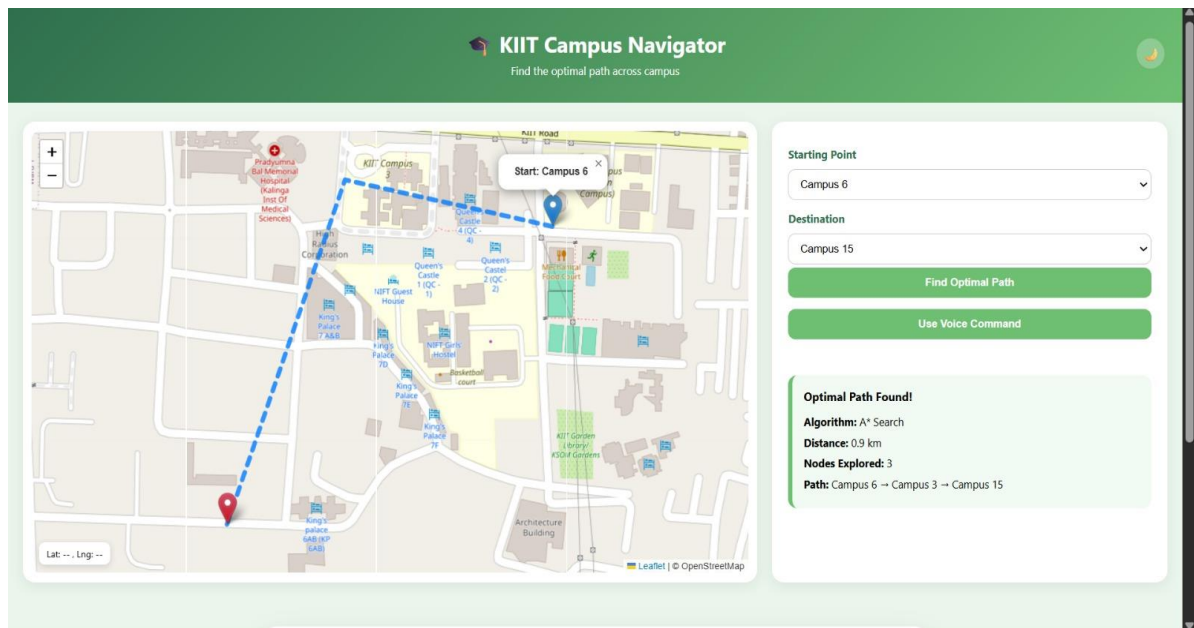


Figure 2: Optimal path between Campus 6 and Campus 15 showing route visualization and calculated distance on the interface.

3. Voice Search Functionality

The user provides input through speech, which is accurately converted into text and used to generate the navigation path. This demonstrates the successful integration of voice recognition with the existing navigation system.

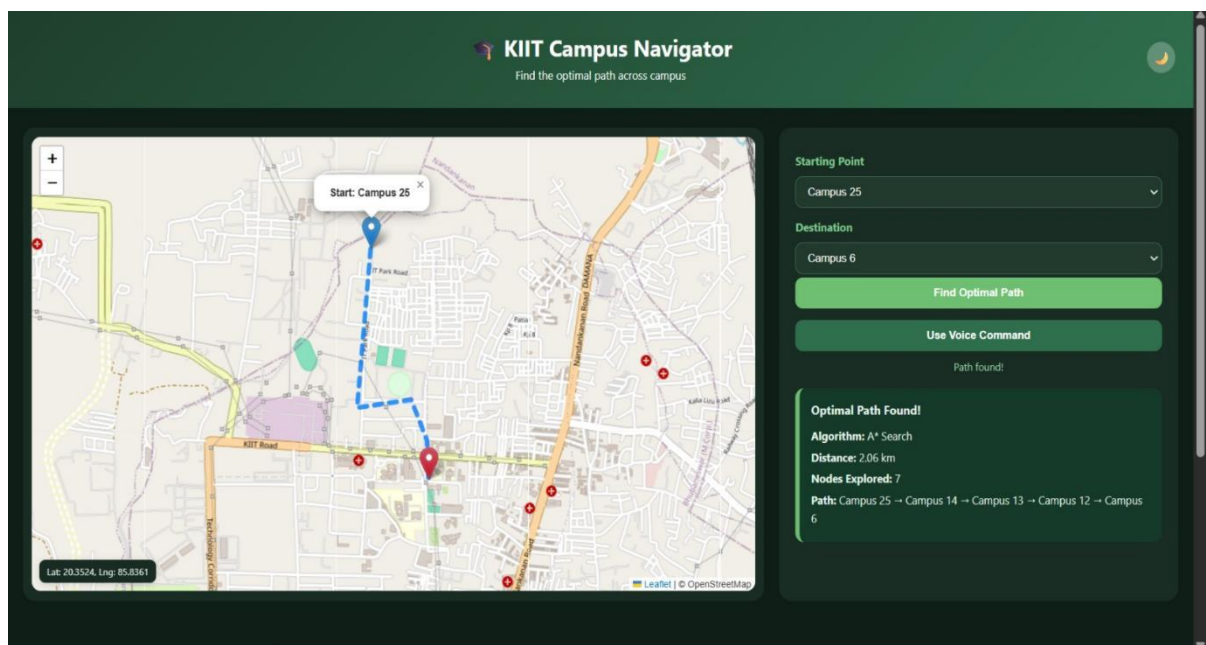


Figure 3: Shortest path generated between Campus 25 and Campus 6 using the A* search algorithm through voice-based input, displayed on the map with route details.

4. Feedback Module

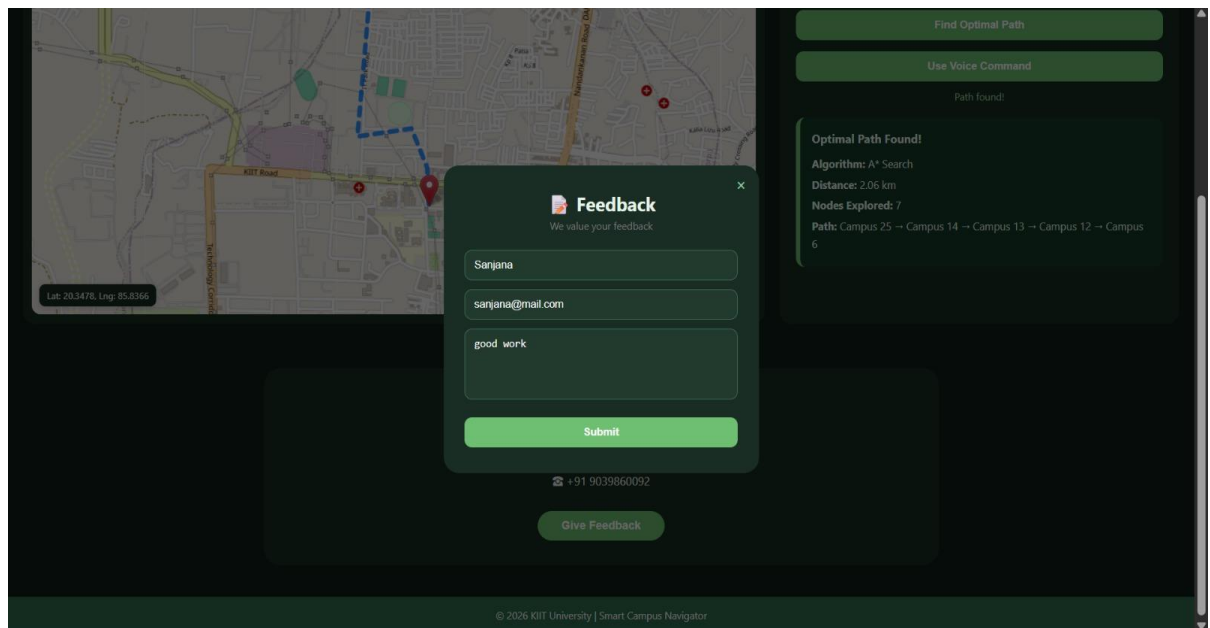


Figure 4: Feedback module interface allowing users to enter their name, email, and suggestions for improving the system.

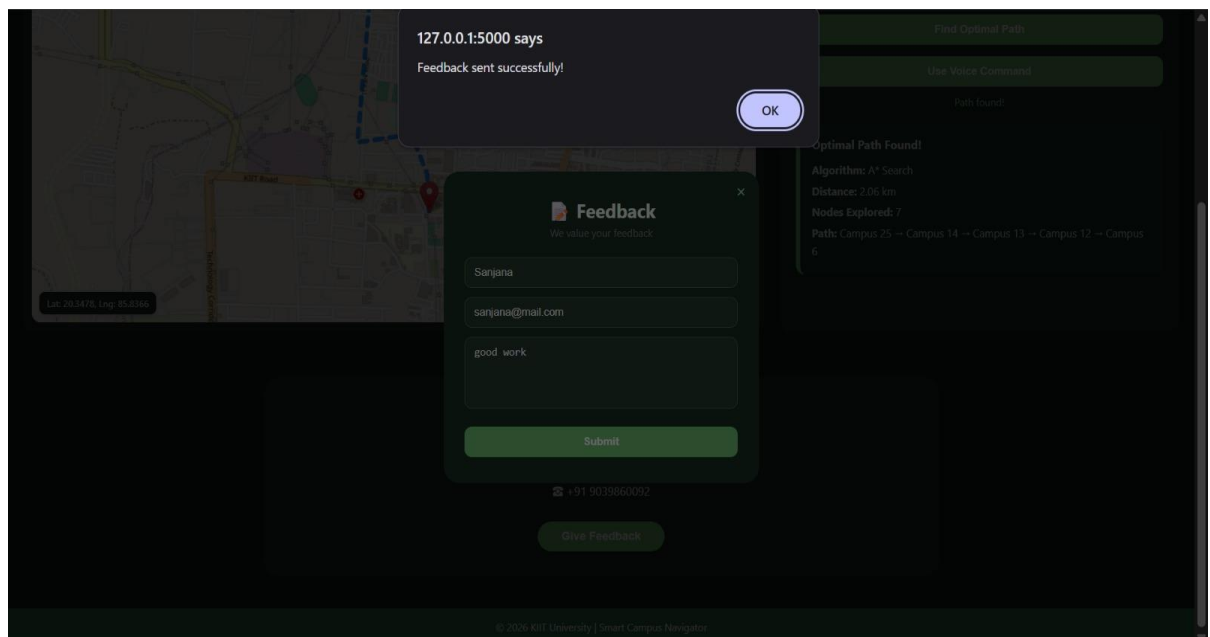


Figure 5: Feedback submission confirmation displayed after successfully submitting user feedback through the system.

The results confirm that the system is capable of providing accurate and efficient navigation within the campus. The addition of voice-based input further enhances usability and makes the system more interactive and convenient for users.

ADVANTAGES

- Provides efficient and accurate navigation within the campus.
- Helps new students and visitors easily locate different campus facilities.
- Uses the A* search algorithm for optimal pathfinding.
- Reduces the time required to find routes between locations.
- User-friendly web interface for easy interaction.
- Supports both text-based and voice-based input for convenience.
- Enables hands-free usage through voice search, improving accessibility.
- Demonstrates practical implementation of Artificial Intelligence in real-world applications.
- Can be extended to develop a smart campus navigation system.

LIMITATIONS

- The system works only for predefined campus locations.
- It does not include real-time traffic or crowd data.
- Navigation is limited to mapped campus areas only.
- The voice recognition feature may be affected by background noise or unclear speech.
- Requires a working microphone for voice input functionality.
- Requires an internet connection to run the web application.
- Newly developed campuses not available on digital maps cannot be included currently.

FUTURE SCOPE

- Integration of real-time navigation with live traffic or crowd data within the campus.
- Addition of GPS-based tracking to provide dynamic route calculation based on the user's current location.
- Development of a mobile application version for improved accessibility and ease of use.
- Enhancement of the voice search feature with better accuracy and noise handling.
- Support for multilingual voice input to make the system usable for a wider range of users.
- Integration of a voice assistant for more interactive and conversational user experience.
- Expansion of the system to include newly developed campuses with updated map data.
- Use of advanced Artificial Intelligence techniques for improved route prediction and personalization.
- Integration with smart campus systems for automated guidance and navigation.

CONCLUSION

The KIIT Campus Navigator successfully demonstrates the practical implementation of Artificial Intelligence in solving real-world navigation problems within a university campus. By utilizing the A* (A-Star) search algorithm, the system efficiently determines the shortest path between different campus locations and provides accurate results through an interactive web interface.

The project effectively combines graph-based representation and heuristic search techniques to ensure optimal pathfinding and improved navigation efficiency. The integration of both text-based and voice-based input enhances the usability of the system, making it more convenient and accessible for users.

The addition of the voice search feature in the updated version further improves user interaction by enabling hands-free navigation. This makes the system more practical for everyday use and highlights the potential of combining AI with user-friendly technologies.

Overall, the project serves as a useful application of Artificial Intelligence concepts and demonstrates how such systems can be extended to develop smarter and more advanced campus navigation solutions in the future.